

Session 7 Quiz - What is Generative AI?

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID :52758]

An AI system that creates new content like text, images, or music is called _____ AI.

- A. Generative
- B. Predictive
- C. Supervised
- D. Unsupervised

Correct Answer: A

Explanation: Generative AI creates new outputs - text, images, audio, video, code. Predictive AI forecasts numeric outcomes. Supervised and unsupervised are training approaches, not output categories.

Q2 [Type: MCQ | Difficulty: Easy | QID :52760]

A program that creates a brand new image from a text description is an image _____ model.

- A. Recognition
- B. Detection
- C. Generation
- D. Classification

Correct Answer: C

Explanation: Image generation creates new images. Recognition, classification, and detection all analyse existing images rather than creating new ones - those are discriminative tasks.

Q3 [Type: MCQ | Difficulty: Easy | QID : 52761]

The instruction or request you give a generative AI to tell it what to create is called a _____.

- A. Label
- B. Feature
- C. Dataset
- D. Prompt

Correct Answer: D

Explanation: A prompt is your direction to the AI - it shapes everything the AI creates. Labels are category tags attached to training data, features are the clues the AI pays attention to, and a dataset is the collection of training examples.

Q4 [Type: MCQ | Difficulty: Medium | QID : 52762]

A specific prompt that includes details like length, style, mood, and subject is likely to produce a better generative AI output than a one-line vague prompt.

- A. True
- B. False

Correct Answer: A (True)

Explanation: Specific details give the AI clearer constraints to remix patterns within - the "you are the director" principle. Detailed instructions narrow down the space of patterns the AI draws from, leading to content that more closely matches your vision. "Make a birthday song" gives the AI almost nothing to work with; "Compose a 30-second upbeat birthday song in pop style with lyrics about cake and friendship" gives it a clear target.

Q5 [Type: MCQ | Difficulty: Easy | QID : 52763]

Generative AI systems truly understand the meaning of the content they create.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Generative AI produces content by matching statistical patterns from training data. It does not understand meaning, hold thoughts, or have intentions. The fluency comes from training on text written by humans who understood - but the AI itself is pattern-matching, not thinking.

Q6 [Type: MCQ | Difficulty: Medium | QID : 52765]

Generative AI has original ideas and that is why it can produce poems and images no human has ever made before.

A. True

B. False

Correct Answer: B (False)

Explanation: New-looking output comes from remixing patterns learned during training, not from original ideas - the AI does not think. Novel-looking output is statistical recombination of patterns the model has seen, applied to a new prompt. The remix artist analogy : a DJ who has studied thousands of songs can produce something that sounds fresh without inventing anything original.

Q7 [Type: MCQ | Difficulty: Easy | QID :52768]

Which trio correctly lists the three main types of generative AI?

A. Text, Image, Audio

B. Cats, Dogs, Birds

C. Classification, Regression, Clustering

D. Email, Search, Maps

Correct Answer: A

Explanation: There are three categories of generative AI: text (ChatGPT, Gemini), image (DALL-E, Midjourney), and audio (Suno, ElevenLabs). B is unrelated. C lists ML task types. D lists app categories.

Q8 [Type: MCQ | Difficulty: Easy | QID :52770]

A common analogy for generative AI is :

A. A detective

B. A calculator

C. A remix artist

D. A traffic light

Correct Answer: C

Explanation: The remix-artist analogy captures how generative AI works: study examples → learn patterns → remix into something new. A is the analogy for discriminative AI. B and D are rule-based examples from earlier topics.

Q9 [Type: MCQ | Difficulty: Medium | QID :52773]

Which task is a job for generative AI?

- A. Writing a funny birthday poem for a friend
- B. Counting how many people are smiling in a party photo
- C. Sorting party photos into folders by who is in them
- D. Detecting whether a comment is positive or negative

Correct Answer: A

Explanation: Generative AI creates new content; the other options all analyse or classify existing content - that is discriminative AI's job. Writing a poem is the only option that produces something new.

Q10 [Type: MCQ | Difficulty: Medium | QID :52778]

Which task does NOT belong in the "Generative AI" category?

- A. Drafting a 4-line poem about your dog
- B. Composing a piano melody in the style of Mozart
- C. Generating a watercolour - style portrait of a friend or family
- D. Counting how many people appear in a group photo

Correct Answer: D

Explanation: Counting people = analysing existing content = discriminative. The other three options all create new content from scratch, which is the defining feature of generative AI.

Q11 [Type: MCQ | Difficulty: Medium | QID :52779]

Which everyday feature is MOST clearly an example of generative AI?

- A. Instagram automatically detecting which friend is in a photo so you can tag them
- B. Gmail's spam filter labelling messages as spam or not spam
- C. Snapchat's "My AI" chatbot writing custom replies to your messages
- D. Face Unlock recognising the owner of the phone

Correct Answer: C

Explanation: Writing custom replies = creating new text = generative. The other three options all classify or recognise existing content (discriminative): tagging friends in photos, labelling spam, recognising a face.

Q12 [Type: MCQ | Difficulty: Medium | QID : 52782]

According to the "Garbage In = Garbage Out" principle, what happens if an AI is trained only on sad poems?

- A. It will automatically learn to write happy poems to balance itself out
- B. It will write poems that have no emotional tone at all
- C. It will struggle to write a happy birthday poem
- D. It will stop working because the data is too biased

Correct Answer: C

Explanation: The quality and variety of the AI's output are strictly limited by the patterns and data present in its training set. A model can only remix what it has been exposed to, so a model fed only sad poems lacks the "happy" patterns needed to produce upbeat output.

Q13 [Type: MCQ | Difficulty: Medium | QID :]

Which of the following is the clearest sign that an AI feature is generative rather than discriminative?

- A. It returns a confidence percentage with each result
- B. It runs in the cloud rather than on your phone
- C. It uses labelled training data
- D. It outputs new content that did not exist before

Correct Answer: D

Explanation: Creating new content is the defining trait of generative AI. A applies to both types of AI. B is unrelated to AI type - both kinds of AI can run anywhere. C also applies to supervised discriminative AI.

Q14 [Type: MCQ | Difficulty: Hard | QID : 52789]

A student copies a full response from an AI chatbot and submits it as their own essay. The teacher discovers this. What is the most accurate framing of what happened?

- A.** The student saved time by using available technology and should face no academic consequences
- B.** The student submitted AI-generated work as their own thinking - a form of academic dishonesty
- C.** The teacher should accept the essay because AI tools are widely available and commonly used now
- D.** The AI is responsible for the dishonesty because it actually wrote the essay, not the student

Correct Answer: B

Explanation: The student misrepresented AI-generated text as their own thinking and analysis. This parallels traditional plagiarism, even though the source is a tool rather than another person. Options A and C dismiss the misrepresentation. Option D shifts responsibility away from the person who made the decision to submit the work.

Q15 [Type: MCQ | Difficulty: Medium | QID :52791]

A friend says: "Generative AI must have its own original ideas - otherwise it could not write a poem I have never seen before." What is the most accurate response?

- A.** Correct - the AI thinks creatively just like humans do and has its own original ideas
- B.** AI generates original ideas only when the user pays for premium access
- C.** The AI invents new words it has never seen before
- D.** New-looking output comes from remixing patterns learned during training

Correct Answer: D

Explanation: Novel-looking output is statistical recombination of patterns the AI has seen, applied to a new prompt - not creative thinking. A is the misconception being tested. C and D fabricate mechanisms that do not exist.

Q16 [Type: MCQ | Difficulty: Hard | QID :52794]

A graphic designer wants to use generative AI to create illustrations "in the style of" a specific working illustrator. What is the most important consideration before doing this?

- A.** The AI tool cannot mimic specific illustrators in any meaningful way, so this idea is not possible
- B.** Speed of generation is the only thing that matters when creating illustrations for clients
- C.** The illustrator's style is their creative work; imitation without consent raises ethical and legal questions
- D.** Once you have purchased the AI tool, no other ethical or legal consideration applies to its outputs

Correct Answer: C

Explanation: An illustrator's style represents years of personal creative work. AI imitation raises real questions about consent, credit, and compensation - these are active legal and ethical debates in the creative industries. Options A, B, and D each dismiss those concerns in different ways.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID :52796]

Scenario: A school wants to build a system that (i) generates personalised birthday cards for students and (ii) checks whether the card images contain anything inappropriate before they are sent.

Question: Which combination of AI types best fits this two-stage system?

- A.** Generative AI for (i); Discriminative AI for (ii)
- B.** Discriminative AI for (i); Generative AI for (ii)
- C.** Generative AI for both stages
- D.** Discriminative AI for both stages

Correct Answer: A

Explanation: Creating cards = generative; checking content for appropriateness = classification = discriminative. The real-world systems often combine the two AI types - each handles a different sub-problem. Option B inverts the assignments. Options C and D collapse the two distinct jobs into one paradigm.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID : 52797]

Scenario: A student uses an image generator that produces text-on-card outputs that often spell words wrong - e.g., "Happy Brithday" instead of "Happy Birthday."

Question: Which statement BEST explains why this happens?

- A.** The model intentionally misspells words as a creative choice to make outputs more unique
- B.** Image generators learn letters as visual shapes, not as spelled words, so letter patterns can come out garbled
- C.** The user must have spelled the prompt wrong when describing the text they wanted on the card
- D.** The model has overfit to the test set and cannot generalise to new text-on-image requests

Correct Answer: B

Explanation: This reinforces the "AI does not understand" theme. Image models treat letters as visual shapes from training images, not as spelled words from a dictionary - so the order and identity of letters can come out wrong even when the overall "look" is right.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID :54180]

Scenario: A student preparing for an exam has access to a generative AI chatbot. Both helpful and harmful ways to use it exist.

Question: Describe one way that would genuinely help them learn, and one way that would prevent learning.

(Respond in 3–4 sentences.)

Model Answer: A helpful use is asking the AI to explain confusing concepts in simpler terms, then attempting practice problems and using the AI to check the reasoning afterward. A harmful use is asking the AI to solve every problem and copying down the answers without working through them. The first builds the student's own thinking; the second replaces it with the AI's output.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 54184]

Scenario: A school district plans to use generative AI to create personalized learning materials for students with diverse needs - different reading levels, learning styles, and language backgrounds. The goal is individualised support at scale.

Question: Identify one significant benefit AND one significant risk, and describe what the district should do to ensure the AI helps rather than harms students.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that AI can adapt materials to each student's reading level and language background - something individual teachers struggle to do at scale. A significant risk is that AI can produce factual errors or culturally insensitive content, and students may not have the background to detect problems with what they receive. To ensure the AI helps rather than harms, the district should have qualified teachers review AI-generated materials before they reach students and gather ongoing feedback from students and families about whether the materials are accurate and useful.